

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash

LLM & Agentic Systems

Agentic systems, LLM agents, tool use, retrieval planning, provider routing/fallback, prompt design, LLM evaluation, OpenAI / Groq APIs, Ollama

Search & Text Mining

RAG, multi-hop QA, multi-view retrieval, RRF, cross-encoder reranking, MMR, entity extraction, web search / scraping, sentence-transformers, spaCy

ML / Deep Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, normalizing flows, diffusion, autodiff, transfer learning

Big Data & Cloud

Apache Spark / PySpark, FastAPI, REST APIs, Docker, AWS (S3, KMS), GCP, Azure, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.
- Participated in a **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and security-finding triage through **AI-assisted workflows** and maintaining a conventions file to keep AI-generated code aligned with team patterns.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input; iterated on prompt design and response-quality heuristics for a real customer-facing experience while building 20+ Node.js + Fastify REST APIs over PostgreSQL.
- Shipped an open-source **LLM auto-commenting feature** in production (4-bit quantized Llama-7B via llama.cpp, per-persona prompts, async queued generation, content-safety filter), built end-to-end with no ML mentor on the team.

Projects

AgenticSearch - Agentic Search & Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables using typed retrieval planning, page rerank, deterministic extraction with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

KG2RAG-Enhanced - Multi-Hop QA & Retrieval

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, PyTorch | UMass CS 685 Advanced NLP (team)

- Led the **multi-view retrieval** component of a team KG-guided RAG pipeline (sub-question decomposition, per-view retrieval, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a **4,905-question** evaluation.
- Ran multi-hop QA experiments over a team knowledge graph of **28,492 entities** on a 4,905-question benchmark, with token/latency logging and 95% binomial confidence intervals on an Ollama + LLaMA-3 pipeline.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an evaluation-improvement agent that proposes new queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible evaluation workflow.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.